

MODIFICATION FOR USE WITH THE CE-150 INTERFACE

This modification lets you use the Software UART while the PC-1500(A) stays docked in the CE-150 printer/cassette interface.

The idea is to use the free (unused) **PB0** input of the LH5811 chip inside the PC-1500(A) instead of **PB2**, which is shared with the cassette input (CMT IN) of the CE-150. The second part of the modification brings the PB0 (input) and PC7 (output) lines and ground out to a connector added to the CE-150 printer/cassette interface.

Procedure for PC-1500(A)

1. Before you start, save your programs (CSAVE): the memory contents may be lost when the batteries are removed. Work on an antistatic surface, because the CMOS chips are sensitive to static electricity. Remove the batteries and undo the screws on the back cover of the PC-1500(A), including those in the battery compartment.
2. Carefully separate the two halves of the case. You will see the board shown in Fig. 1. With a thin wire, connect the contact of the 60-pin side connector (pin 9, see the photo) to the pad on the PCB next to the LH5811 chip (or LH5810) as shown on Fig. 2 and Fig. 3. For protection it is best to add a 1 k Ω series resistor. Pin 9 of the side connector is now connected to the **PB0** pin of the LH5811. *Nota bene: I have heard that some PC-1500(A) units already have this connection from the factory (according to the Technical Reference Manual it depends on the production date). I own about 10 PC-1500 and PC-1500A units, but none of them has it.*

With the pull-up in place you can check the connection in RUN mode: PRINT (PEEK# &F00F) AND 1 gives 1, and 0 while pin 9 is connected to GND.

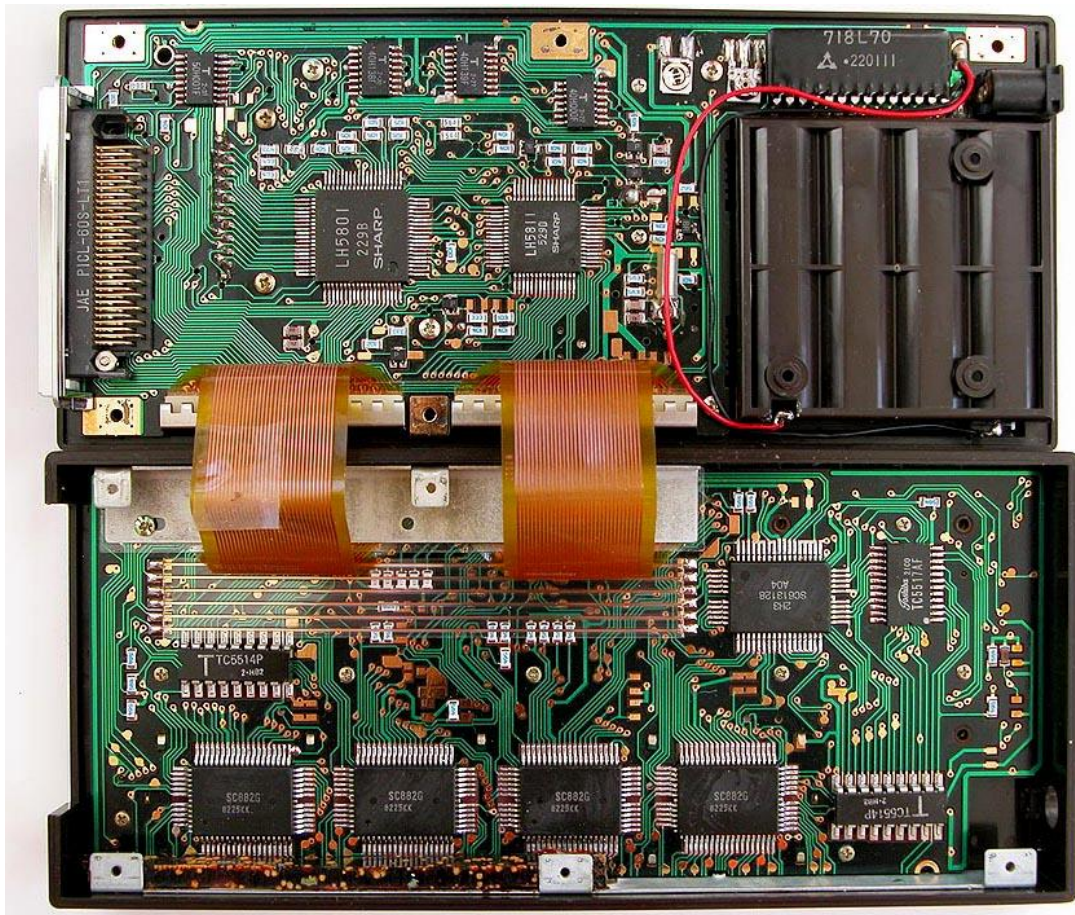


Fig. 1 Interior of a PC-1500A

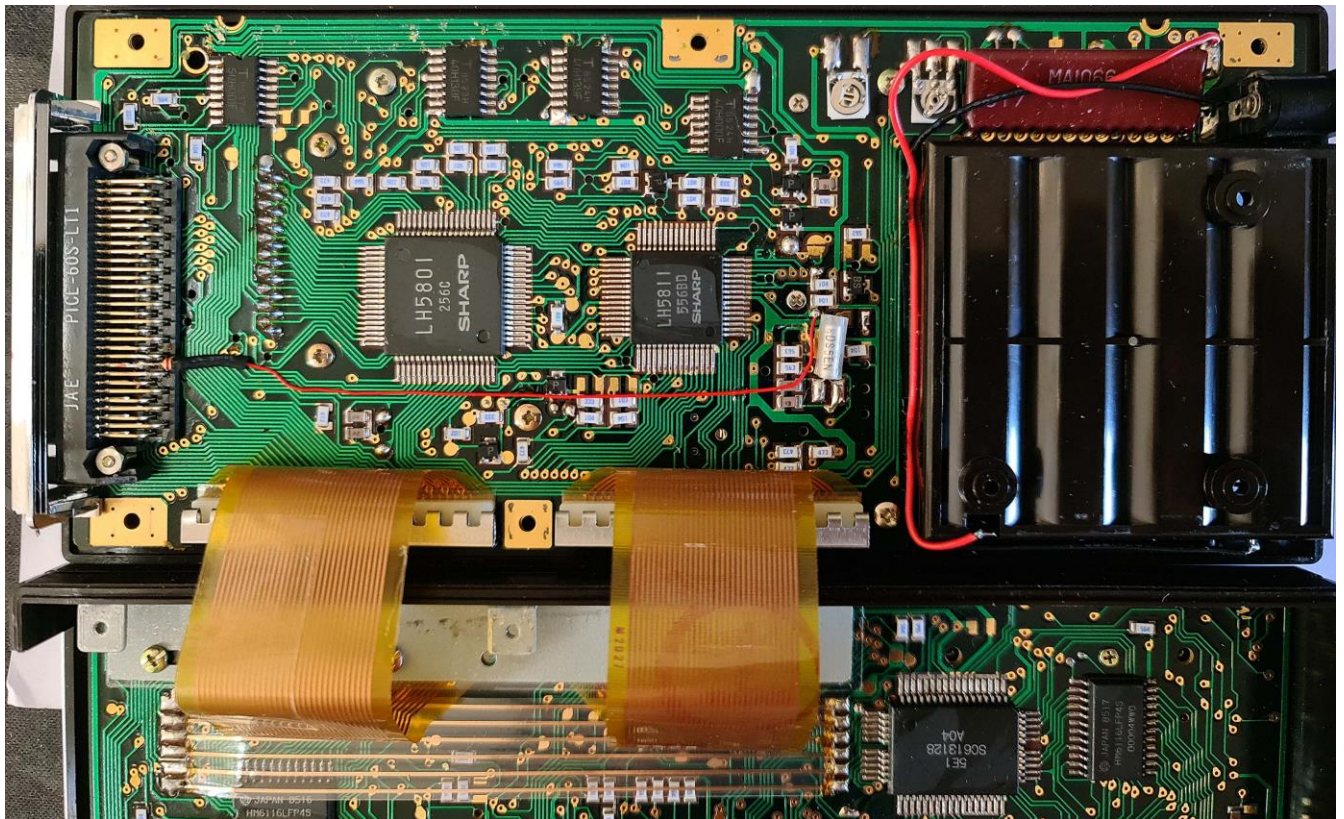


Fig. 2 Connection to PB0 implemented (thin red wire)

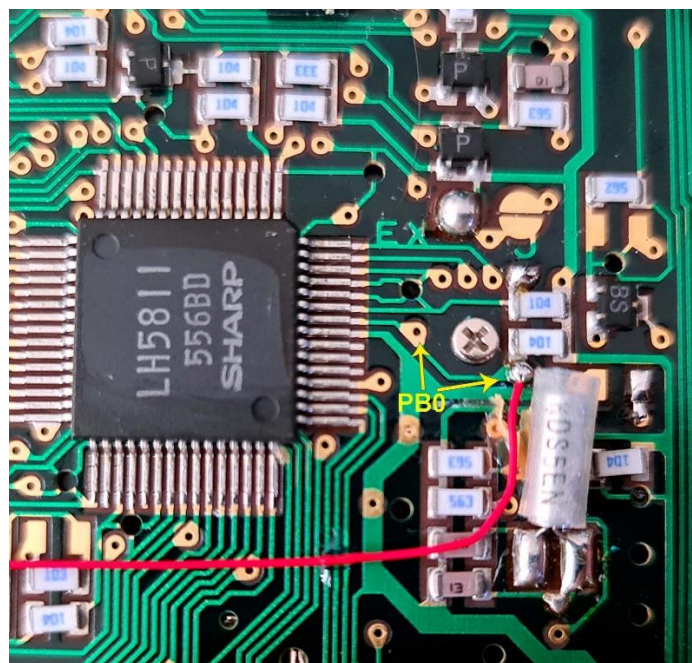


Fig. 3 Close-up of the PB0 connection point

Procedure for CE-150 cassette/printer interface

1. Now open the CE-150 (first remove the PC-1500 and disconnect the AC adapter). Undo the 8 screws on the back of the case, then gently prise the two halves apart at the joint; they separate easily. Inside you will see the CE-150 board (Fig. 4), which you can clean if necessary. It is also worth replacing the old, probably worn-out NiCd battery pack with a pack of two 18650 Li-ion cells, which must have a 2S BMS protection board (as shown, see Fig. 4).

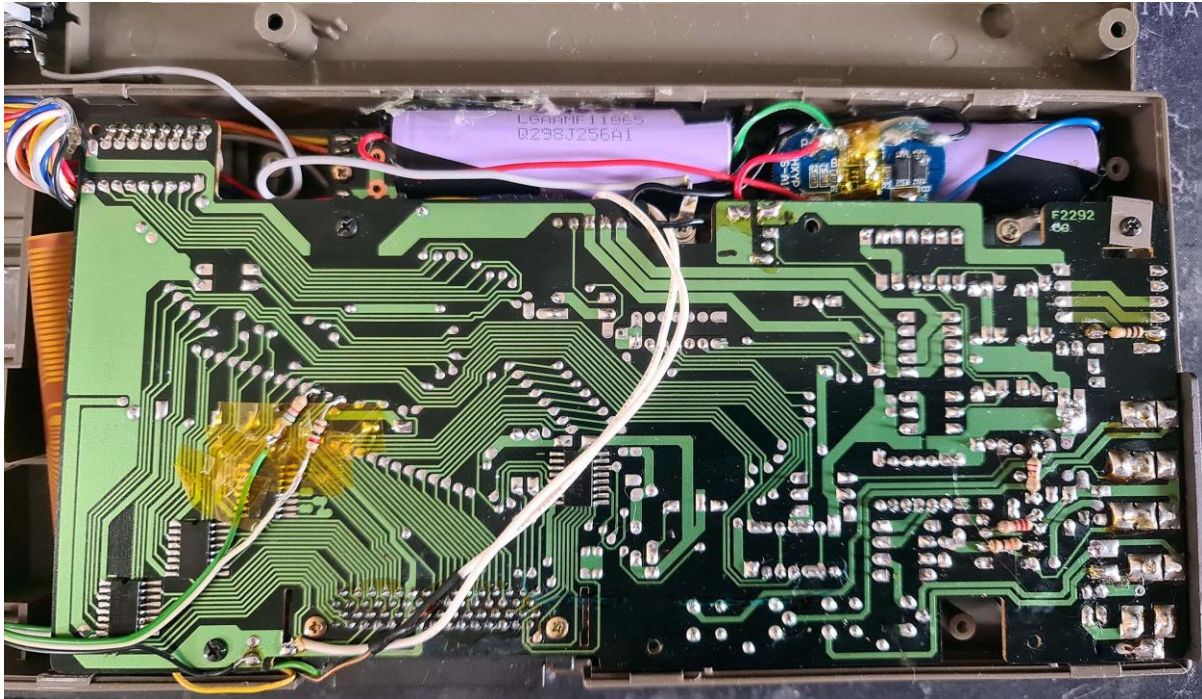


Fig. 4 Inside the CE-150 printer/cassette interface: cable and resistors soldered to the PCB

2. Solder a 3-core cable to: (1) ground (GND) on the PCB, (2) the PB0 point through a 1–10 k Ω resistor, (3) the PC7 point through a 1–5 k Ω resistor. The points are shown in Fig. 5. Solder the other end of the cable to a socket, e.g. a 3.5 mm stereo jack, and mount it in a free spot on the case (as shown, see Fig. 6). Write down which contact is which, for example: tip = TX (PC7), ring = RX (PB0), sleeve = GND. You can also simply bring longer wires out of the interface without a socket: less elegant, but it works just as well.

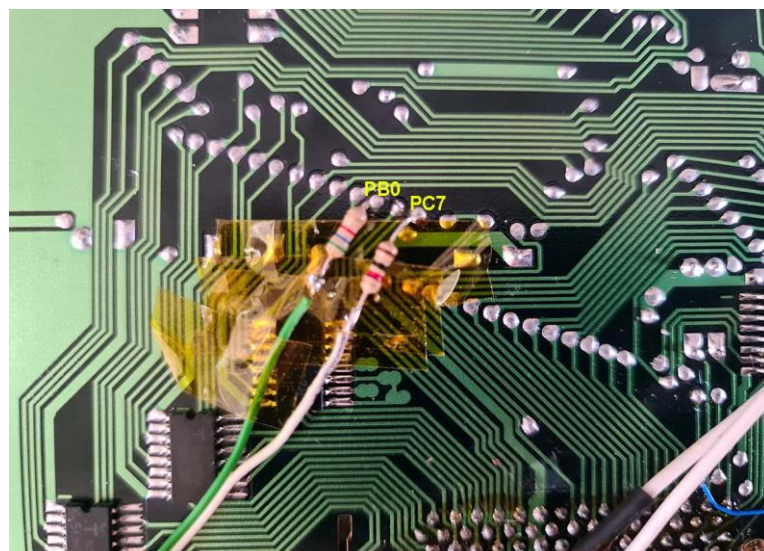


Fig. 5 Close-up of the wires and resistors soldered to the PB0 and PC7 points

3. Besides the jack, Fig. 6 shows a red-green LED that indicates the state of the RX and TX lines. It is optional. If you add it, use a high-efficiency LED, so that a very small current already makes the red and green parts glow without loading the port lines too much. I used an RGB LED (blue part unused) with 3.3 k Ω resistors; it glows clearly at about 0.5-1.0 mA. *Note: The LEDs will be lit in the idle state (the UART idle level is 'high').*

4.

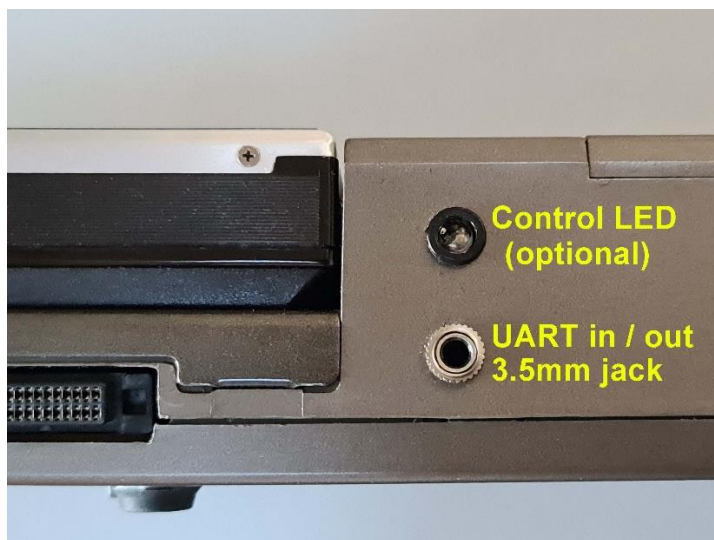


Fig.6 UART jack on the CE-150's rear panel and the optional LED

5. Run the installer and answer the RX PORT question with 0 (PB0). The Software UART is now available on the CE-150, and you can use both the UART and the cassette recorder, because they now use separate inputs! Of course not at the same time, and not while printing or doing other work. Receiving/sending through the Software UART, loading/saving programs, printing and calculations all use the full power of the LH5801 CPU, which cannot multitask... but it gives a lot of other fun!

Good luck!